Static Timing Analysis Report

Post-Route — All PVT Corners

VSDBabySoC on Sky130 FD SC HD

NAME: B. KEERTHAN STANLEY

ROLL NO: 24EG104C03

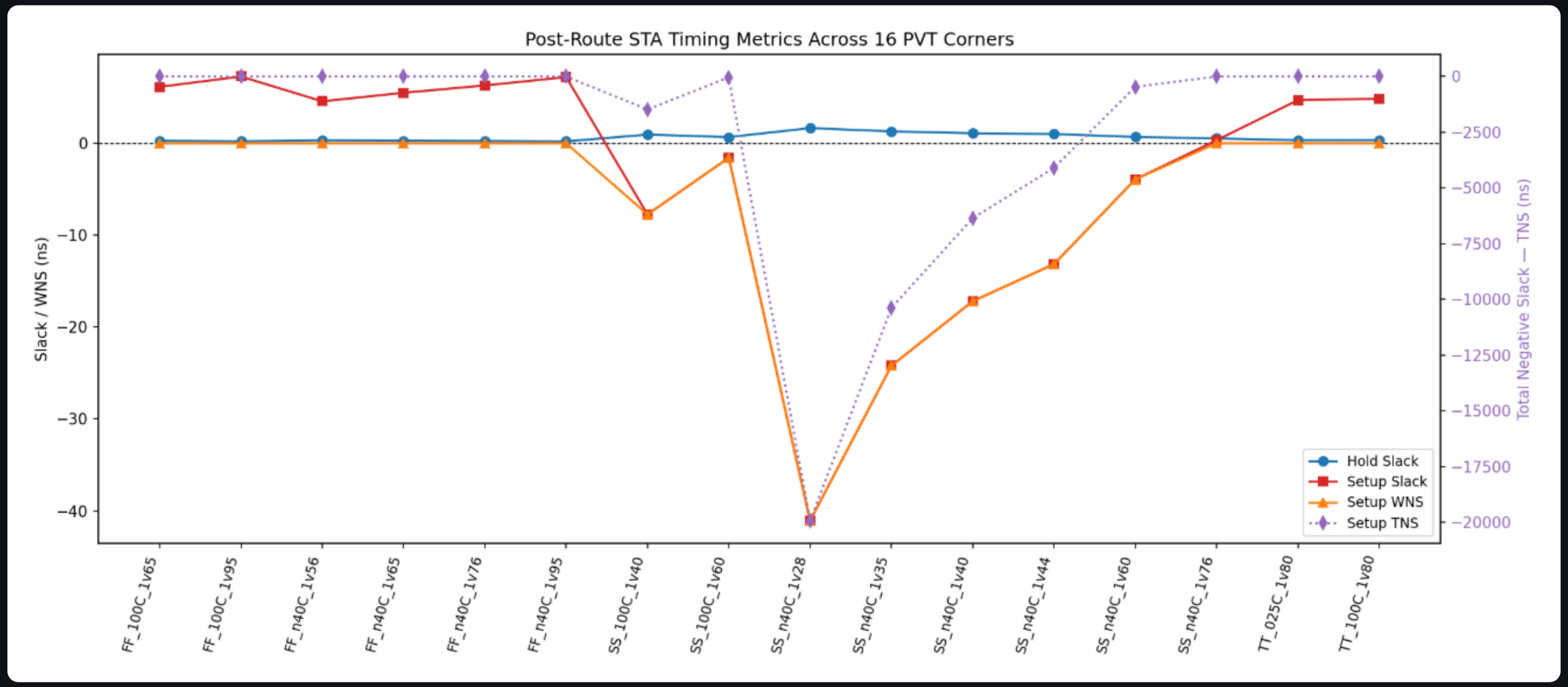
Corner Explanation Table

The 16 PVT corners used in the sweep

Corner	Process	Voltage	Temp	Meaning
ff_100C_1v65	Fast-Fast	1.65 V	100 °C	Fast, hot, low V
ff_100C_1v95	Fast-Fast	1.95 V	100 °C	Fast, hot, high V
ff_n40C_1v56	Fast-Fast	1.56 V	-40 °C	Fast, cold, very low V
ff_n40C_1v65	Fast-Fast	1.65 V	-40 °C	Fast, cold, low V
ff_n40C_1v76	Fast-Fast	1.76 V	-40 °C	Fast, cold, mid V
ff_n40C_1v95	Fast-Fast	1.95 V	-40 °C	Worst case for hold
ss_100C_1v40	Slow-Slow	1.40 V	100 °C	Slow, hot, low V
ss_100C_1v60	Slow-Slow	1.60 V	100 °C	Slow, hot, nominal V
ss_n40C_1v28	Slow-Slow	1.28 V	-40 °C	Worst case for setup
ss_n40C_1v35	Slow-Slow	1.35 V	-40 °C	Slow, cold, low V
ss_n40C_1v40	Slow-Slow	1.40 V	-40 °C	Slow, cold, low V
ss_n40C_1v44	Slow-Slow	1.44 V	-40 °C	Slow, cold, low V
ss_n40C_1v60	Slow-Slow	1.60 V	-40 °C	Slow, cold, nominal V
ss_n40C_1v76	Slow-Slow	1.76 V	-40 °C	Slow, cold, high V
tt_025C_1v80	Typical	1.80 V	25 °C	Nominal baseline
tt_100C_1v80	Typical	1.80 V	100 °C	Hot typical

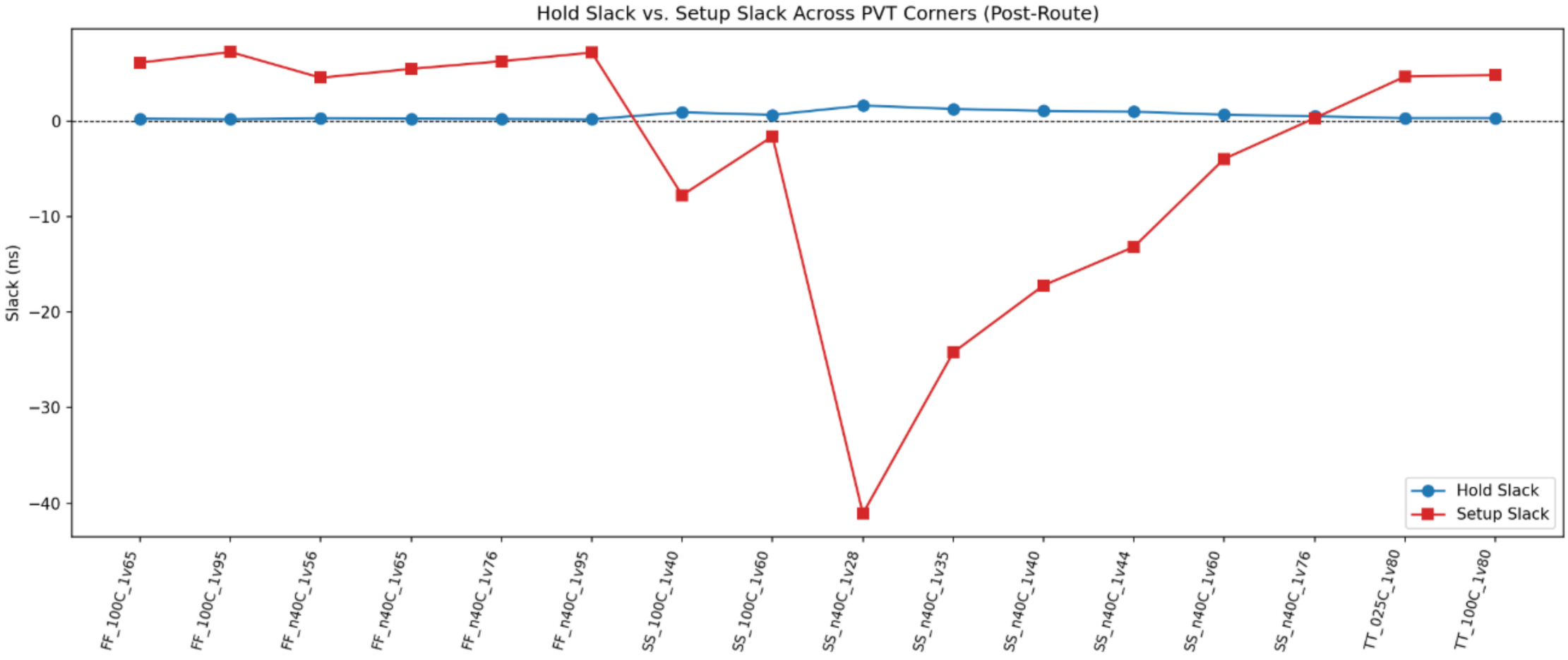
Line Graph

Setup / hold metrics across all 16 PVT corners (post-route)



Line Graph

Hold slack vs. setup slack across PVT corners (post-route)



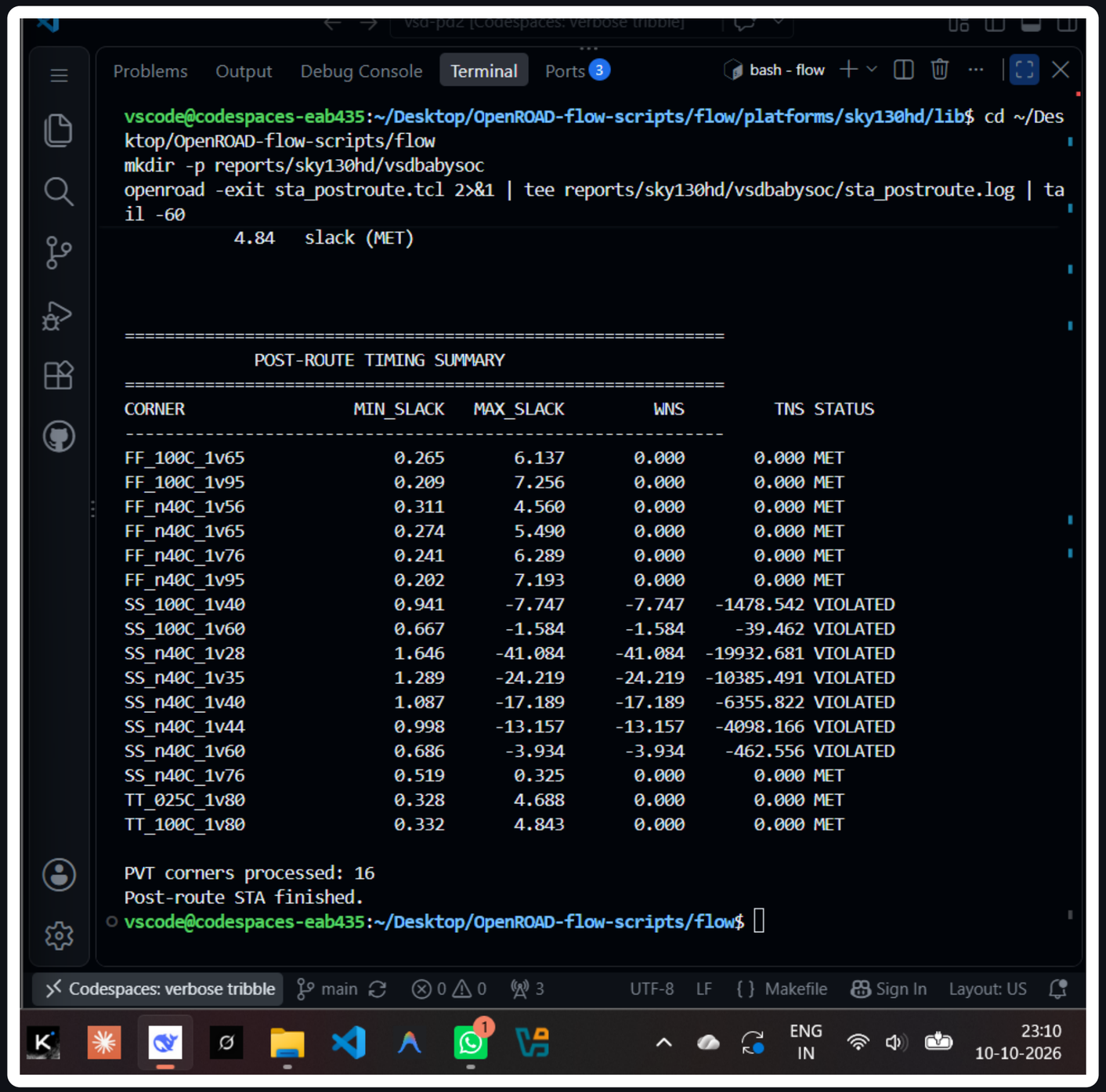
WNS / TNS Summary

Setup WNS / TNS and slack at each PVT corner (ns)

Corner	Setup WNS	Setup TNS	Hold Slack	Setup Slack	Setup
ff_100C_1v65	0.00	0.00	0.265	6.137	PASS
ff_100C_1v95	0.00	0.00	0.209	7.256	PASS
ff_n40C_1v56	0.00	0.00	0.311	4.560	PASS
ff_n40C_1v65	0.00	0.00	0.274	5.490	PASS
ff_n40C_1v76	0.00	0.00	0.241	6.289	PASS
ff_n40C_1v95	0.00	0.00	0.202	7.193	PASS
ss_100C_1v40	-7.75	-1478.54	0.941	-7.747	FAIL
ss_100C_1v60	-1.58	-39.46	0.667	-1.584	FAIL
ss_n40C_1v28	-41.08	-19932.68	1.646	-41.084	FAIL
ss_n40C_1v35	-24.22	-10385.49	1.289	-24.219	FAIL
ss_n40C_1v40	-17.19	-6355.82	1.087	-17.189	FAIL
ss_n40C_1v44	-13.16	-4098.17	0.998	-13.157	FAIL
ss_n40C_1v60	-3.93	-462.56	0.686	-3.934	FAIL
ss_n40C_1v76	0.00	0.00	0.519	0.325	PASS
tt_025C_1v80	0.00	0.00	0.328	4.688	PASS
tt_100C_1v80	0.00	0.00	0.332	4.843	PASS

Post-Route Timing Summary

Same table captured in the terminal after running sta_postroute.tcl



Post-Route Timing Summary

Third capture of the summary table (lower resolution)

```
============================================================
          POST-ROUTE TIMING SUMMARY
============================================================
CORNER                  MIN_SLACK   MAX_SLACK         WNS         TNS STATUS
------------------------------------------------------------
FF_100C_1v65                0.265       6.137       0.000       0.000 MET
FF_100C_1v95                0.209       7.256       0.000       0.000 MET
FF_n40C_1v56                0.311       4.560       0.000       0.000 MET
FF_n40C_1v65                0.274       5.490       0.000       0.000 MET
FF_n40C_1v76                0.241       6.289       0.000       0.000 MET
FF_n40C_1v95                0.202       7.193       0.000       0.000 MET
SS_100C_1v40                0.941      -7.747      -7.747   -1478.542 VIOLATED
SS_100C_1v60                0.667      -1.584      -1.584     -39.462 VIOLATED
SS_n40C_1v28                1.646     -41.084     -41.084  -19932.681 VIOLATED
SS_n40C_1v35                1.289     -24.219     -24.219  -10385.491 VIOLATED
SS_n40C_1v40                1.087     -17.189     -17.189   -6355.822 VIOLATED
SS_n40C_1v44                0.998     -13.157     -13.157   -4098.166 VIOLATED
SS_n40C_1v60                0.686      -3.934      -3.934    -462.556 VIOLATED
SS_n40C_1v76                0.519       0.325       0.000       0.000 MET
TT_025C_1v80                0.328       4.688       0.000       0.000 MET
TT_100C_1v80                0.332       4.843       0.000       0.000 MET

PVT corners processed: 16
Post-route STA finished.
```

Evidence

Run command and the tail of the last corner's path report (tt_100C_1v80)

```
vscode@codespaces-eab435:~/Desktop/OpenROAD-flow-scripts/flow/platforms/sky130hd/lib$ cd ~/Des
ktop/OpenROAD-flow-scripts/flow
mkdir -p reports/sky130hd/vsdbabysoc
openroad -exit sta_postroute.tcl 2>&1 | tee reports/sky130hd/vsdbabysoc/sta_postroute.log | ta
il -60
-----------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.19    1.19   clock network delay (propagated)
   0.00    1.19 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.85    2.05 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.61    2.65 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.81    3.46 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.78    4.24 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.33    4.57 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.48    5.05 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.31    5.36 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.88    6.24 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.45    6.69 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.33    7.03 v place504/X (sky130_fd_sc_hd__buf_4)
   0.25    7.27 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    7.27 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           7.27   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.21   12.21   clock network delay (propagated)
   0.00   12.21   clock reconvergence pessimism
          12.21 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.09   12.12   library setup time
          12.12   data required time
-----------------------------------------------------------
          12.12   data required time
          -7.27   data arrival time
-----------------------------------------------------------
           4.84   slack (MET)
```

Pre-layout vs Post-route

Setup and hold slack per corner (ns); pre-layout values from the earlier STA report

Corner	Setup pre	Setup post	Change	Hold pre	Hold post	Post status
ff_100C_1v65	4.055	6.137	+2.082	0.249	0.265	MET
ff_100C_1v95	5.567	7.256	+1.689	0.196	0.209	MET
ff_n40C_1v56	2.278	4.560	+2.282	0.292	0.311	MET
ff_n40C_1v65	3.411	5.490	+2.079	0.255	0.274	MET
ff_n40C_1v76	4.430	6.289	+1.859	0.224	0.241	MET
ff_n40C_1v95	5.592	7.193	+1.601	0.187	0.202	MET
ss_100C_1v40	-17.122	-7.747	+9.375	0.905	0.941	VIOLATED
ss_100C_1v60	-7.874	-1.584	+6.290	0.642	0.667	VIOLATED
ss_n40C_1v28	-62.563	-41.084	+21.479	1.830	1.646	VIOLATED
ss_n40C_1v35	-39.697	-24.219	+15.478	1.348	1.289	VIOLATED
ss_n40C_1v40	-29.694	-17.189	+12.505	1.125	1.087	VIOLATED
ss_n40C_1v44	-23.908	-13.157	+10.751	0.991	0.998	VIOLATED
ss_n40C_1v60	-10.609	-3.934	+6.675	0.663	0.686	VIOLATED
ss_n40C_1v76	-4.382	0.325	+4.707	0.504	0.519	MET
tt_025C_1v80	1.939	4.688	+2.749	0.310	0.328	MET
tt_100C_1v80	2.094	4.843	+2.749	0.314	0.332	MET

Not a like-for-like comparison: the pre-layout run used standalone OpenSTA 2.4.0 on the synthesized netlist; the post-route run uses OpenROAD STA with propagated clocks on the routed design.

Corner 1/16 — ff_100C_1v65

Fast, hot, low V

PVT CONDITIONS

Process
Fast-Fast

Corner
ff_100C_1v65

Voltage
1.65 V

Temperature
100 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
6.137

Hold Slack (ns)
0.265

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 5.85 ns Required 11.99 ns Clock 11.00 ns

```
Running STA: FF_100C_1v65
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_100C_1v65

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.05    1.05   clock network delay (propagated)
   0.00    1.05 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.73    1.78 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.50    2.27 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    2.95 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.53    3.48 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.25    3.73 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.41    4.14 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.18    4.32 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.72    5.04 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.32    5.36 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.28    5.64 v place504/X (sky130_fd_sc_hd__buf_4)
   0.21    5.85 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    5.85 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           5.85   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.06   12.06   clock network delay (propagated)
   0.00   12.06   clock reconvergence pessimism
          12.06 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.07   11.99   library setup time
          11.99   data required time
---------------------------------------------------------
          11.99   data required time
          -5.85   data arrival time
---------------------------------------------------------
           6.14   slack (MET)
```

Corner 2/16 — ff_100C_1v95

Fast, hot, high V

PVT CONDITIONS

Process
Fast-Fast

Corner
ff_100C_1v95

Voltage
1.95 V

Temperature
100 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
7.256

Hold Slack (ns)
0.209

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 4.57 ns Required 11.82 ns Clock 11.00 ns

```
Running STA: FF_100C_1v95
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
         (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_100C_1v95

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.86    0.86   clock network delay (propagated)
   0.00    0.86 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.59    1.44 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.38    1.82 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.53    2.35 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.40    2.75 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.20    2.95 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.31    3.26 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.13    3.39 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.56    3.95 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.23    4.18 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.22    4.41 v place504/X (sky130_fd_sc_hd__buf_4)
   0.16    4.57 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    4.57 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           4.57   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.87   11.87   clock network delay (propagated)
   0.00   11.87   clock reconvergence pessimism
          11.87 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.04   11.82   library setup time
          11.82   data required time
---------------------------------------------------------
          11.82   data required time
          -4.57   data arrival time
---------------------------------------------------------
           7.26   slack (MET)
```

Corner 3/16 — ff_n40C_1v56

Fast, cold, very low V

PVT CONDITIONS

Process
Fast-Fast

Corner
ff_n40C_1v56

Voltage
1.56 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
4.560

Hold Slack (ns)
0.311

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 7.68 ns Required 12.24 ns Clock 11.00 ns

```
Running STA: FF_n40C_1v56
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v56

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.32    1.32   clock network delay (propagated)
   0.00    1.32 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.96    2.27 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.57    2.85 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   1.02    3.86 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.60    4.47 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.39    4.86 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.60    5.46 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.23    5.69 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.96    6.65 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.38    7.02 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.36    7.38 v place504/X (sky130_fd_sc_hd__buf_4)
   0.30    7.68 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    7.68 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           7.68   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.34   12.34   clock network delay (propagated)
   0.00   12.34   clock reconvergence pessimism
          12.34 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.10   12.24   library setup time
          12.24   data required time
---------------------------------------------------------
          12.24   data required time
          -7.68   data arrival time
---------------------------------------------------------
           4.56   slack (MET)
```

Corner 4/16 — ff_n40C_1v65

Fast, cold, low V

PVT CONDITIONS

Process
Fast-Fast

Corner
ff_n40C_1v65

Voltage
1.65 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
5.490

Hold Slack (ns)
0.274

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 6.61 ns Required 12.10 ns Clock 11.00 ns

```
Running STA: FF_n40C_1v65
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
         (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v65

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.16    1.16   clock network delay (propagated)
   0.00    1.16 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.84    2.00 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.49    2.50 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.87    3.36 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.50    3.87 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.33    4.19 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.51    4.71 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.18    4.89 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.83    5.72 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.32    6.04 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.31    6.35 v place504/X (sky130_fd_sc_hd__buf_4)
   0.26    6.61 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    6.61 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           6.61   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.18   12.18   clock network delay (propagated)
   0.00   12.18   clock reconvergence pessimism
          12.18 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08   12.10   library setup time
          12.10   data required time
---------------------------------------------------------
          12.10   data required time
          -6.61   data arrival time
---------------------------------------------------------
           5.49   slack (MET)
```

Corner 5/16 — ff_n40C_1v76

Fast, cold, mid V

PVT CONDITIONS

Process
Fast-Fast

Corner
ff_n40C_1v76

Voltage
1.76 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
6.289

Hold Slack (ns)
0.241

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 5.69 ns Required 11.98 ns Clock 11.00 ns

```
Running STA: FF_n40C_1v76
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v76

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.02    1.02   clock network delay (propagated)
   0.00    1.02 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.74    1.77 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.42    2.18 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.74    2.92 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.43    3.35 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.27    3.62 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.44    4.06 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.15    4.20 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.72    4.92 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.26    5.19 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.28    5.47 v place504/X (sky130_fd_sc_hd__buf_4)
   0.22    5.69 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    5.69 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           5.69   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.04   12.04   clock network delay (propagated)
   0.00   12.04   clock reconvergence pessimism
          12.04 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.06   11.98   library setup time
          11.98   data required time
---------------------------------------------------------
          11.98   data required time
          -5.69   data arrival time
---------------------------------------------------------
           6.29   slack (MET)
```

Corner 6/16 — ff_n40C_1v95

Worst case for hold

PVT CONDITIONS

Process
Fast-Fast

Corner
ff_n40C_1v95

Voltage
1.95 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
7.193

Hold Slack (ns)
0.202

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 4.64 ns Required 11.84 ns Clock 11.00 ns

```
Running STA: FF_n40C_1v95
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v95

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.86    0.86   clock network delay (propagated)
   0.00    0.86 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.63    1.49 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.34    1.83 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.59    2.42 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.35    2.76 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.21    2.98 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.35    3.33 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.11    3.43 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.59    4.02 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.20    4.23 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.24    4.46 v place504/X (sky130_fd_sc_hd__buf_4)
   0.18    4.64 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    4.64 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           4.64   data arrival time

  11.00   11.00   clock clk (rise edge)
   0.88   11.88   clock network delay (propagated)
   0.00   11.88   clock reconvergence pessimism
          11.88 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.04   11.84   library setup time
          11.84   data required time
---------------------------------------------------------
          11.84   data required time
          -4.64   data arrival time
---------------------------------------------------------
           7.19   slack (MET)
```

Corner 7/16 — ss_100C_1v40

Slow, hot, low V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_100C_1v40

Voltage
1.40 V

Temperature
100 °C

TIMING RESULTS

Setup WNS (ns)
-7.747

Setup TNS (ns)
-1478.542

Setup Slack (ns)
-7.747

Hold Slack (ns)
0.941

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 21.35 ns Required 13.60 ns Clock 11.00 ns

```
Running STA: SS_100C_1v40
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_100C_1v40

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.89    2.89   clock network delay (propagated)
   0.00    2.89 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.21    5.10 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.98    7.09 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   2.32    9.40 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   2.61   12.02 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   1.19   13.21 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   1.32   14.53 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   1.02   15.55 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   2.50   18.05 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   1.71   19.76 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.92   20.68 v place504/X (sky130_fd_sc_hd__buf_4)
   0.67   21.35 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00   21.35 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          21.35   data arrival time

  11.00   11.00   clock clk (rise edge)
   2.93   13.93   clock network delay (propagated)
   0.00   13.93   clock reconvergence pessimism
          13.93 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.34   13.60   library setup time
          13.60   data required time
---------------------------------------------------------
          13.60   data required time
         -21.35   data arrival time
---------------------------------------------------------
          -7.75   slack (VIOLATED)
```

Corner 8/16 — ss_100C_1v60

Slow, hot, nominal V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_100C_1v60

Voltage
1.60 V

Temperature
100 °C

TIMING RESULTS

Setup WNS (ns)
-1.584

Setup TNS (ns)
-39.462

Setup Slack (ns)
-1.584

Hold Slack (ns)
0.667

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 14.49 ns Required 12.90 ns Clock 11.00 ns

```
---------------------------------------------------------
Running STA: SS_100C_1v60
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_100C_1v60

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.08    2.08   clock network delay (propagated)
   0.00    2.08 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.56    3.65 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.32    4.97 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   1.57    6.54 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   1.72    8.26 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.76    9.02 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.91    9.92 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.67   10.60 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   1.71   12.31 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   1.07   13.38 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.64   14.03 v place504/X (sky130_fd_sc_hd__buf_4)
   0.46   14.49 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00   14.49 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.49   data arrival time

  11.00   11.00   clock clk (rise edge)
   2.11   13.11   clock network delay (propagated)
   0.00   13.11   clock reconvergence pessimism
          13.11 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.21   12.90   library setup time
          12.90   data required time
---------------------------------------------------------
          12.90   data required time
         -14.49   data arrival time
---------------------------------------------------------
          -1.58   slack (VIOLATED)
```

Corner 9/16 — ss_n40C_1v28

Worst case for setup

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_n40C_1v28

Voltage
1.28 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
-41.084

Setup TNS (ns)
-19932.681

Setup Slack (ns)
-41.084

Hold Slack (ns)
1.646

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[12]$_DFF_P_

To: core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_

Data arrival 58.40 ns Required 17.32 ns Clock 11.00 ns

```
Running STA: SS_n40C_1v28
Startpoint: core.CPU_src1_value_a3[12]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v28

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   8.00    8.00   clock network delay (propagated)
   0.00    8.00 ^ core.CPU_src1_value_a3[12]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   5.15   13.14 ^ core.CPU_src1_value_a3[12]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   4.05   17.20 v _10830_/SUM (sky130_fd_sc_hd__ha_1)
   5.85   23.04 ^ _07812_/Y (sky130_fd_sc_hd__nor4_1)
   2.80   25.84 v _07813_/Y (sky130_fd_sc_hd__nand2_1)
   5.26   31.10 ^ _07814_/Y (sky130_fd_sc_hd__a211oi_1)
   2.52   33.62 v _08170_/Y (sky130_fd_sc_hd__nor2_1)
   4.62   38.24 ^ _08174_/Y (sky130_fd_sc_hd__a21oi_1)
   2.70   40.94 v _08325_/Y (sky130_fd_sc_hd__nand2_1)
   1.67   42.61 v _08329_/X (sky130_fd_sc_hd__and3_1)
   6.48   49.09 ^ _08333_/Y (sky130_fd_sc_hd__o32ai_1)
   4.26   53.34 ^ place518/X (sky130_fd_sc_hd__buf_4)
   1.66   55.00 v _08357_/Y (sky130_fd_sc_hd__nand2_1)
   3.40   58.40 ^ _09226_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   58.40 ^ core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          58.40   data arrival time

  11.00   11.00   clock clk (rise edge)
   7.55   18.55   clock network delay (propagated)
   0.00   18.55   clock reconvergence pessimism
          18.55 ^ core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.23   17.32   library setup time
          17.32   data required time
---------------------------------------------------------
          17.32   data required time
         -58.40   data arrival time
---------------------------------------------------------
         -41.08   slack (VIOLATED)
```

Corner 10/16 — ss_n40C_1v35

Slow, cold, low V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_n40C_1v35

Voltage
1.35 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
-24.219

Setup TNS (ns)
-10385.491

Setup Slack (ns)
-24.219

Hold Slack (ns)
1.289

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[12]$_DFF_P_

To: core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_

Data arrival 39.77 ns Required 15.55 ns Clock 11.00 ns

```
Running STA: SS_n40C_1v35
Startpoint: core.CPU_src1_value_a3[12]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_
         (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v35

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   5.60    5.60   clock network delay (propagated)
   0.00    5.60 ^ core.CPU_src1_value_a3[12]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   3.68    9.29 ^ core.CPU_src1_value_a3[12]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   2.88   12.16 v _10830_/SUM (sky130_fd_sc_hd__ha_1)
   4.01   16.17 ^ _07812_/Y (sky130_fd_sc_hd__nor4_1)
   1.74   17.91 v _07813_/Y (sky130_fd_sc_hd__nand2_1)
   3.59   21.50 ^ _07814_/Y (sky130_fd_sc_hd__a211oi_1)
   1.55   23.05 v _08170_/Y (sky130_fd_sc_hd__nor2_1)
   3.25   26.30 ^ _08174_/Y (sky130_fd_sc_hd__a21oi_1)
   1.73   28.04 v _08325_/Y (sky130_fd_sc_hd__nand2_1)
   1.16   29.20 v _08329_/X (sky130_fd_sc_hd__and3_1)
   4.29   33.49 ^ _08333_/Y (sky130_fd_sc_hd__o32ai_1)
   2.58   36.07 ^ place518/X (sky130_fd_sc_hd__buf_4)
   1.28   37.36 v _08357_/Y (sky130_fd_sc_hd__nand2_1)
   2.41   39.77 ^ _09226_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   39.77 ^ core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          39.77   data arrival time

  11.00   11.00   clock clk (rise edge)
   5.30   16.30   clock network delay (propagated)
   0.00   16.30   clock reconvergence pessimism
          16.30 ^ core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.75   15.55   library setup time
          15.55   data required time
---------------------------------------------------------
          15.55   data required time
         -39.77   data arrival time
---------------------------------------------------------
         -24.22   slack (VIOLATED)
```

Corner 11/16 — ss_n40C_1v40

Slow, cold, low V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_n40C_1v40

Voltage
1.40 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
-17.189

Setup TNS (ns)
-6355.822

Setup Slack (ns)
-17.189

Hold Slack (ns)
1.087

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[12]$_DFF_P_

To: core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_

Data arrival 31.93 ns Required 14.74 ns Clock 11.00 ns

```
Running STA: SS_n40C_1v40
Startpoint: core.CPU_src1_value_a3[12]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v40

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   4.56    4.56   clock network delay (propagated)
   0.00    4.56 ^ core.CPU_src1_value_a3[12]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   3.02    7.59 ^ core.CPU_src1_value_a3[12]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   2.33    9.92 v _10830_/SUM (sky130_fd_sc_hd__ha_1)
   3.20   13.12 ^ _07812_/Y (sky130_fd_sc_hd__nor4_1)
   1.35   14.47 v _07813_/Y (sky130_fd_sc_hd__nand2_1)
   2.87   17.34 ^ _07814_/Y (sky130_fd_sc_hd__a211oi_1)
   1.20   18.53 v _08170_/Y (sky130_fd_sc_hd__nor2_1)
   2.66   21.19 ^ _08174_/Y (sky130_fd_sc_hd__a21oi_1)
   1.37   22.56 v _08325_/Y (sky130_fd_sc_hd__nand2_1)
   0.93   23.49 v _08329_/X (sky130_fd_sc_hd__and3_1)
   3.37   26.87 ^ _08333_/Y (sky130_fd_sc_hd__o32ai_1)
   2.00   28.86 ^ place518/X (sky130_fd_sc_hd__buf_4)
   1.10   29.96 v _08357_/Y (sky130_fd_sc_hd__nand2_1)
   1.97   31.93 ^ _09226_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   31.93 ^ core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          31.93   data arrival time

  11.00   11.00   clock clk (rise edge)
   4.32   15.32   clock network delay (propagated)
   0.00   15.32   clock reconvergence pessimism
          15.32 ^ core.CPU_Xreg_value_a4[27][27]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.58   14.74   library setup time
          14.74   data required time
---------------------------------------------------------
          14.74   data required time
         -31.93   data arrival time
---------------------------------------------------------
         -17.19   slack (VIOLATED)
```

Corner 12/16 — ss_n40C_1v44

Slow, cold, low V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_n40C_1v44

Voltage
1.44 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
-13.157

Setup TNS (ns)
-4098.166

Setup Slack (ns)
-13.157

Hold Slack (ns)
0.998

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 27.61 ns Required 14.45 ns Clock 11.00 ns

```
Running STA: SS_n40C_1v44
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v44

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   3.87    3.87   clock network delay (propagated)
   0.00    3.87 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.97    6.84 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   2.25    9.09 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   3.55   12.64 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   2.90   15.54 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   1.43   16.97 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   2.04   19.01 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   1.30   20.31 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   3.30   23.61 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   2.00   25.61 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   1.05   26.66 v place504/X (sky130_fd_sc_hd__buf_4)
   0.95   27.61 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00   27.61 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          27.61   data arrival time

  11.00   11.00   clock clk (rise edge)
   3.93   14.93   clock network delay (propagated)
   0.00   14.93   clock reconvergence pessimism
          14.93 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.48   14.45   library setup time
          14.45   data required time
---------------------------------------------------------
          14.45   data required time
         -27.61   data arrival time
---------------------------------------------------------
         -13.16   slack (VIOLATED)
```

Corner 13/16 — ss_n40C_1v60

Slow, cold, nominal V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_n40C_1v60

Voltage
1.60 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
-3.934

Setup TNS (ns)
-462.556

Setup Slack (ns)
-3.934

Hold Slack (ns)
0.686

STATUS

Setup: **VIOLATED** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 17.25 ns Required 13.32 ns Clock 11.00 ns

```
Running STA: SS_n40C_1v60
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v60

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   2.54    2.54   clock network delay (propagated)
   0.00    2.54 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.94    4.48 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.43    5.91 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   2.16    8.07 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   1.73    9.80 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.92   10.72 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   1.25   11.96 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.77   12.73 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   2.06   14.79 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   1.16   15.94 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.71   16.65 v place504/X (sky130_fd_sc_hd__buf_4)
   0.60   17.25 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00   17.25 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          17.25   data arrival time

  11.00   11.00   clock clk (rise edge)
   2.58   13.58   clock network delay (propagated)
   0.00   13.58   clock reconvergence pessimism
          13.58 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.26   13.32   library setup time
          13.32   data required time
---------------------------------------------------------
          13.32   data required time
         -17.25   data arrival time
---------------------------------------------------------
          -3.93   slack (VIOLATED)
```

Corner 14/16 — ss_n40C_1v76

Slow, cold, high V

PVT CONDITIONS

Process
Slow-Slow

Corner
ss_n40C_1v76

Voltage
1.76 V

Temperature
-40 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
0.325

Hold Slack (ns)
0.519

STATUS

Setup: **MET** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 12.41 ns Required 12.74 ns Clock 11.00 ns

```
Running STA: SS_n40C_1v76
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v76

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.91    1.91   clock network delay (propagated)
   0.00    1.91 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.43    3.34 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.01    4.35 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   1.52    5.87 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   1.23    7.10 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.65    7.75 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.88    8.63 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.53    9.16 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   1.47   10.62 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.81   11.44 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.54   11.98 v place504/X (sky130_fd_sc_hd__buf_4)
   0.43   12.41 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00   12.41 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          12.41   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.94   12.94   clock network delay (propagated)
   0.00   12.94   clock reconvergence pessimism
          12.94 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.20   12.74   library setup time
          12.74   data required time
---------------------------------------------------------
          12.74   data required time
         -12.41   data arrival time
---------------------------------------------------------
           0.33   slack (MET)
```

Corner 15/16 — tt_025C_1v80

Nominal baseline

PVT CONDITIONS

Process
Typical

Corner
tt_025C_1v80

Voltage
1.80 V

Temperature
25 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
4.688

Hold Slack (ns)
0.328

STATUS

Setup: **MET**

Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 7.47 ns Required 12.15 ns Clock 11.00 ns

```
-------------------------------------------------------------
Running STA: TT_025C_1v80
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
         (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: TT_025C_1v80

  Delay    Time   Description
-------------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.23    1.23   clock network delay (propagated)
   0.00    1.23 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.89    2.12 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.59    2.72 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.88    3.59 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.73    4.32 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.37    4.69 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.52    5.21 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.30    5.51 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.91    6.42 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.44    6.86 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.35    7.20 v place504/X (sky130_fd_sc_hd__buf_4)
   0.26    7.47 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    7.47 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           7.47   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.25   12.25   clock network delay (propagated)
   0.00   12.25   clock reconvergence pessimism
          12.25 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.10   12.15   library setup time
          12.15   data required time
-------------------------------------------------------------
          12.15   data required time
          -7.47   data arrival time
-------------------------------------------------------------
           4.69   slack (MET)
```

Corner 16/16 — tt_100C_1v80

Hot typical

PVT CONDITIONS

Process
Typical

Corner
tt_100C_1v80

Voltage
1.80 V

Temperature
100 °C

TIMING RESULTS

Setup WNS (ns)
0.000

Setup TNS (ns)
0.000

Setup Slack (ns)
4.843

Hold Slack (ns)
0.332

STATUS

Setup: **MET** Hold: **MET**

WORST SETUP PATH

From: core.CPU_src1_value_a3[18]$_DFF_P_

To: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_

Data arrival 7.27 ns Required 12.12 ns Clock 11.00 ns

```
Running STA: TT_100C_1v80
Startpoint: core.CPU_src1_value_a3[18]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_
         (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: TT_100C_1v80

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   1.19    1.19   clock network delay (propagated)
   0.00    1.19 ^ core.CPU_src1_value_a3[18]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.85    2.05 ^ core.CPU_src1_value_a3[18]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.61    2.65 v _10849_/SUM (sky130_fd_sc_hd__ha_1)
   0.81    3.46 ^ _07965_/Y (sky130_fd_sc_hd__nor4_1)
   0.78    4.24 v _08104_/Y (sky130_fd_sc_hd__o2111ai_1)
   0.33    4.57 v _08261_/X (sky130_fd_sc_hd__a21o_1)
   0.48    5.05 ^ _08271_/Y (sky130_fd_sc_hd__a211oi_1)
   0.31    5.36 v _08272_/Y (sky130_fd_sc_hd__o311ai_0)
   0.88    6.24 ^ _08276_/Y (sky130_fd_sc_hd__o21ai_0)
   0.45    6.69 v _08277_/Y (sky130_fd_sc_hd__mux2i_2)
   0.33    7.03 v place504/X (sky130_fd_sc_hd__buf_4)
   0.25    7.27 ^ _08949_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    7.27 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           7.27   data arrival time

  11.00   11.00   clock clk (rise edge)
   1.21   12.21   clock network delay (propagated)
   0.00   12.21   clock reconvergence pessimism
          12.21 ^ core.CPU_Xreg_value_a4[17][25]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.09   12.12   library setup time
          12.12   data required time
---------------------------------------------------------
          12.12   data required time
          -7.27   data arrival time
---------------------------------------------------------
           4.84   slack (MET)
```

Analysis & Conclusion

Interpretation of the post-route corner sweep, not just the raw numbers

Overall trend

Post-route STA ran on VSDBabySoC across all 16 sky130_fd_sc_hd PVT corners at an 11.00 ns clock period. All six ff_* corners, both tt_* corners and ss_n40C_1v76 meet setup. The other seven ss_* corners violate setup. Hold slack is positive at every corner, so no hold violations.

Why ss_n40C_1v28 is worst

The worst setup violation is ss_n40C_1v28: WNS -41.084 ns, TNS -19932.681 ns. It combines slow-slow process, the lowest supply in the sweep (1.28 V) and -40 °C. The worst path (CPU_src1_value_a3[12] to CPU_Xreg_value_a4[27][27]) has a 58.40 ns arrival against a 17.32 ns required time, and the propagated clock latency alone is about 8 ns at this corner.

Voltage trend at -40 °C

Setup slack improves steadily as supply rises: -41.08 (1.28 V), -24.22 (1.35 V), -17.19 (1.40 V), -13.16 (1.44 V), -3.93 (1.60 V), then +0.33 ns at 1.76 V, which just meets timing. At 100 °C the same trend holds: -7.75 ns at 1.40 V and -1.58 ns at 1.60 V.

Setup vs. hold

Setup is bounded by the clock period, so slow cells push data arrival past the required time. Hold only needs delay above a small margin, so slower cells help it: hold slack is largest at ss_n40C_1v28 (1.646 ns) while setup fails worst there. The tightest hold corner is ff_n40C_1v95 at +0.202 ns.

Versus the pre-layout report

The pass/fail pattern is the same, with ff and tt passing and ss failing, except ss_n40C_1v76, which now meets setup (+0.325 ns, pre-layout -4.382 ns). Every failing ss corner has a smaller violation than before (worst -62.56 to -41.08 ns). The two runs used different tools and netlists, so this is a trend comparison rather than an exact one.

Practical implication

The design meets setup at the nominal (tt) and fast corners. Closing the ss_n40C and ss_100C corners would need repair_timing or cell upsizing on the CPU_Xreg and CPU_src1 paths, a faster clock tree, or a longer clock period for those corners. Whether the very low-voltage corners (1.28 to 1.44 V) are required for signoff depends on the target operating spec.